Introduction to Biomaterials

Lecture [4]

Phase Diagrams, Crystal Structures, and Metallic Biomaterials

[Metallic Biomaterials]

Goals

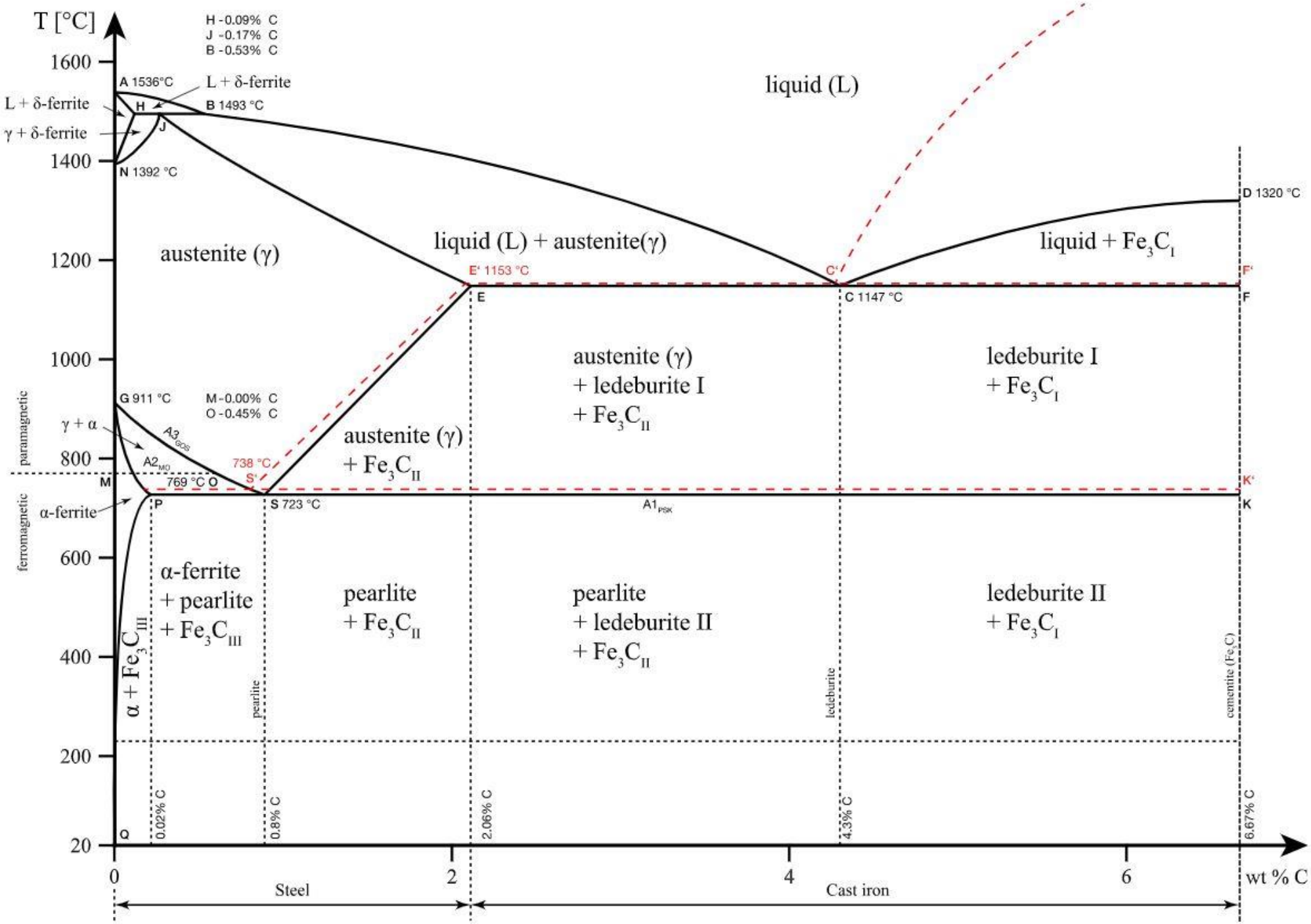
- Understand the basic role of phase diagrams in predicting metal phases and microstructures.
- Recognize the main crystal structures in metals: FCC, BCC, and HCP.
- Explain why metals are widely used in biomedical implants, devices, and surgical tools.
- Identify the main metallic biomaterials: titanium, stainless steel, Co–Cr alloys, nitinol, tantalum, and magnesium.
- Relate metal properties such as corrosion resistance, strength, wear resistance, elastic modulus, and biocompatibility to biomedical applications.
- Compare metallic biomaterials based on their advantages, limitations, and clinical uses.

Phase Diagrams

Phase Diagrams

- **Phase diagrams** are graphical representations of the phases present in an alloy at different conditions of temperature, pressure, or chemical composition.

- **Iron-Carbon system (**Fe-C) diagram helps engineers understand how carbon and temperature change iron structure and properties. This knowledge is used to design steels for buildings, machines, tools, and medical devices.


T [°C]
H - 0.09% C
J - 0.17% C
B - 0.53% C
A 1536°C
L + δ-ferrite
B 1493 °C
L + δ-ferrite
H
γ + δ-ferrite
J
N 1392 °C
liquid (L)
D 1320 °C
austenite (γ)
liquid (L) + austenite(γ)
liquid + Fe3CI
E' 1153 °C
C'
F'
E
C 1147 °C
F
austenite (γ)
+ ledeburite I
+ Fe3CII
ledeburite I
+ Fe3CI
G 911 °C
M - 0.00% C
O - 0.45% C
γ + α
A3GOS
A2MO
738 °C
S'
austenite (γ)
+ Fe3CII
paramagnetic
M
769 °C O
K'
α-ferrite
P
S 723 °C
A1PSK
K
ferromagnetic
α-ferrite
+ pearlite
+ Fe3CIII
pearlite
+ Fe3CII
pearlite
+ ledeburite II
+ Fe3CII
ledeburite II
+ Fe3CI
α + Fe3CIII
pearlite
ledeburite
cementite (Fe3C)
Q
0.02% C
0.8% C
2.06% C
4.3% C
6.67% C
1600
1400
1200
1000
800
600
400
200
20
0
2
4
6
wt % C
Steel
Cast iron

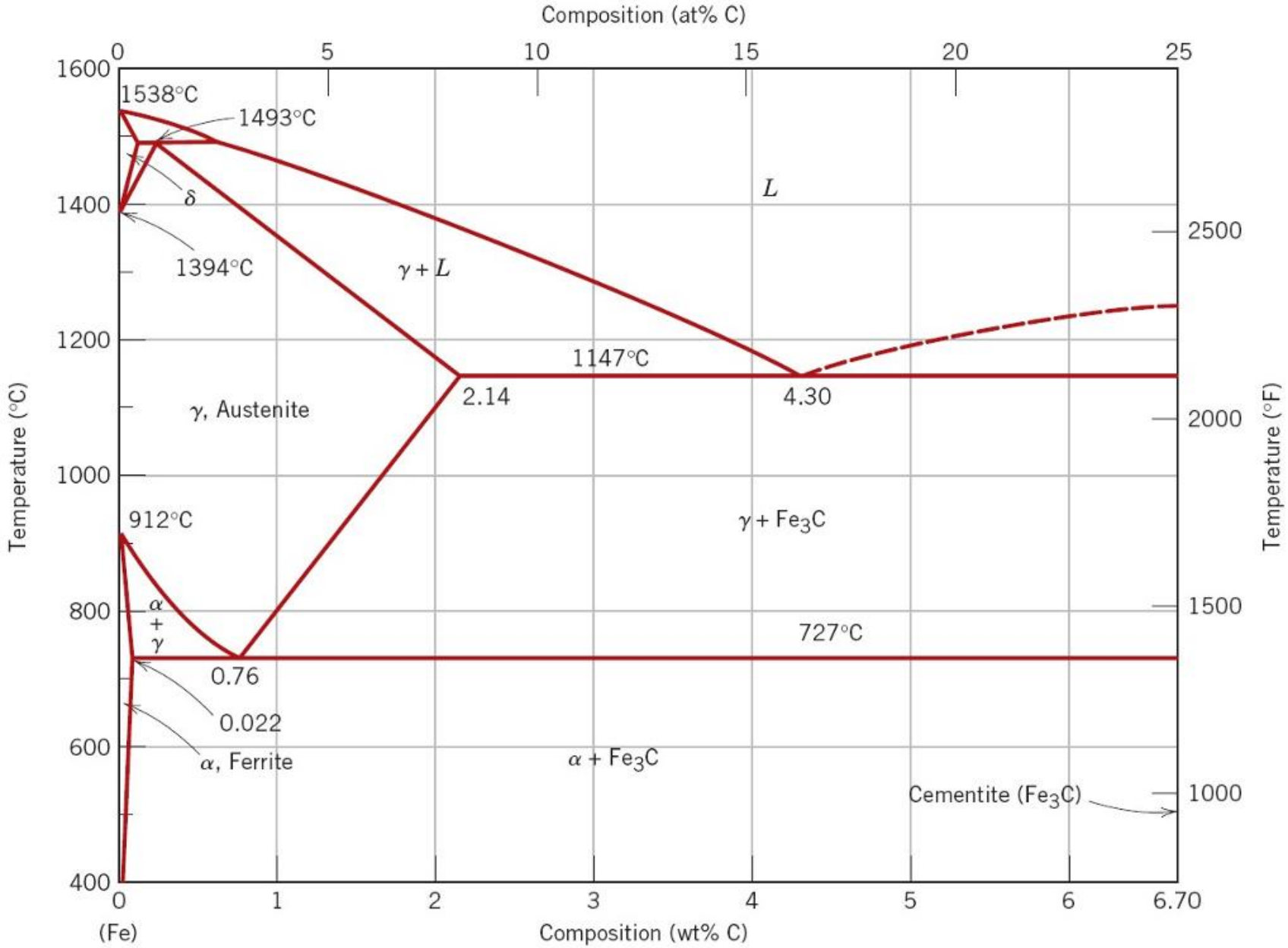


Figure 9.24 The iron–iron carbide phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

Main reference for review

- https://www.youtube.com/watch?v=HdlCehVdNc4

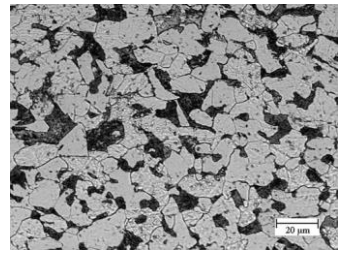
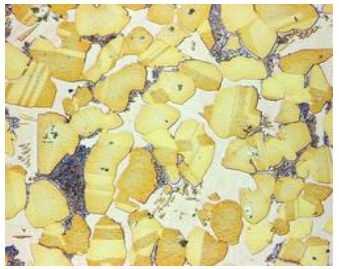
- Phase diagrams can be made for many alloy systems, but the Iron-Carbon diagram is the most famous and widely taught because steel and cast iron are the backbone of modern industry.
- Phase diagrams exist for many systems:
  - Cu–Zn (brass)
  - Cu–Ni
  - Ti–Al
  - Ni–Cr
  - Ag–Cu
  - Mg alloys
  - Biomaterial alloys like Ti–6Al–4V

Main reference for review

- Phase diagrams exist for many systems:
  - Cu–Zn (Copper–Zinc phase diagram / Brass phase diagram)
  - Cu–Ni (Copper–Nickel phase diagram)
  - Ti–Al (Titanium–Aluminum phase diagram)
  - Ni–Cr (Nickel–Chromium phase diagram)
  - Ag–Cu (Silver–Copper phase diagram)
  - Mg alloys (Magnesium–Aluminum ,Magnesium–Zinc , Magnesium–Calcium phase diagram)
  - Biomaterial alloys like Ti–6Al–4V (Titanium Grade 5) (Titanium–Aluminum–Vanadium phase diagram)

What is this graph?

- A **binary phase diagram** showing what structure forms in **iron + carbon alloys** at different:
  - Temperature (vertical axis)
  - Carbon % (horizontal axis)
  - It predicts what happens during heating/cooling.
- **Why important?**
  - Because steel properties depend on microstructure, not only chemistry.


refers to the internal structure and arrangement of a material's components, which is crucial for designing engineering structures with enhanced performances and specific mechanical behaviors.

Why Iron is Important in Materials

- Explain simply:
  - Most used engineering metal in the world
  - Strong, cheap, recyclable
- Used in:
  - Buildings
  - Cars
  - Machines
  - Surgical tools
  - Stainless steel implants
- Why Carbon is added?
  - Pure iron is soft.
  - Adding carbon changes hardness, strength, wear resistance.



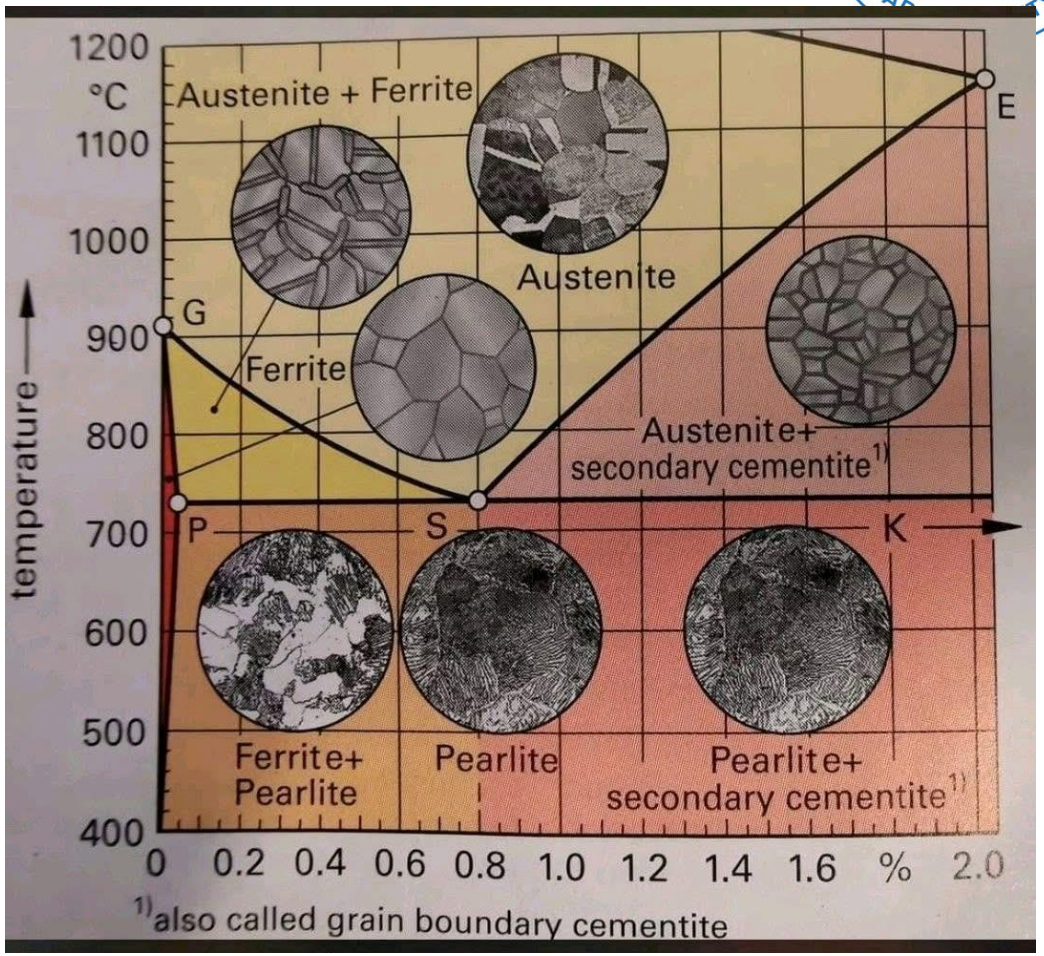
Difference Between Steel and Cast Iron

- Use this only:
  - **Steel** = low carbon (0–2.1%)
  - **Cast iron** = high carbon (2.1–6.67%)

- **Meaning:**
  - Steel = stronger + workable
  - Cast iron = hard + brittle + easy to cast

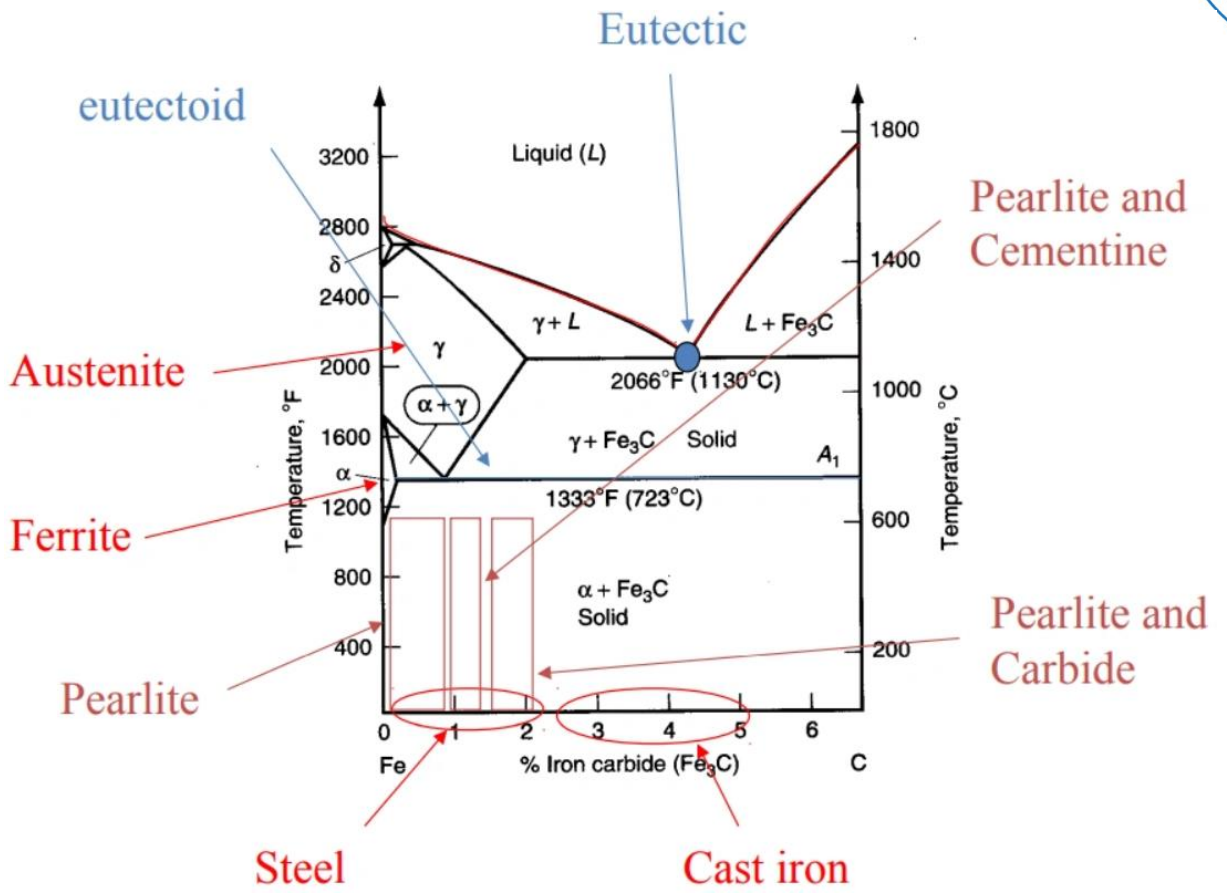
Main Phases Must Know

- Only these four:
- Ferrite (α)
  - Soft
  - Ductile
  - Low carbon solubility
- Austenite (γ)
  - Exists at high temperature
  - Important in heat treatment
- Cementite (Fe_3C)
  - Very hard
  - Brittle
- Pearlite
  - Mixture of ferrite + cementite
  - Balanced strength/hardness

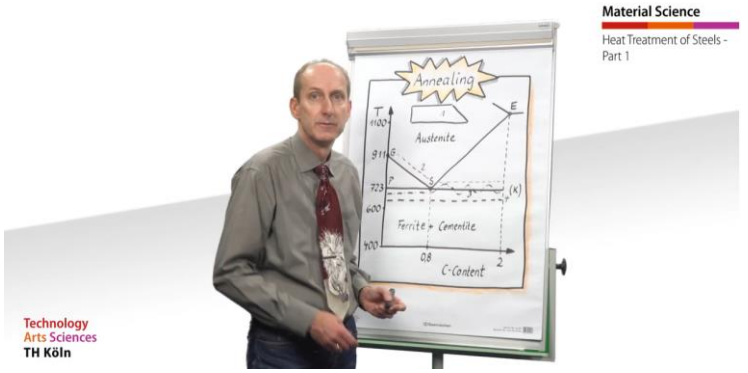


Main Phases Must Know

- Only these four:
- Ferrite (α)
  - Soft
  - Ductile
  - Low carbon solubility
- Austenite (γ)
  - Exists at high temperature
  - Important in heat treatment
- Cementite (Fe_3C)
  - Very hard
  - Brittle
- Pearlite
  - Mixture of ferrite + cementite
  - Balanced strength/hardness



Why Heating/Cooling Matters



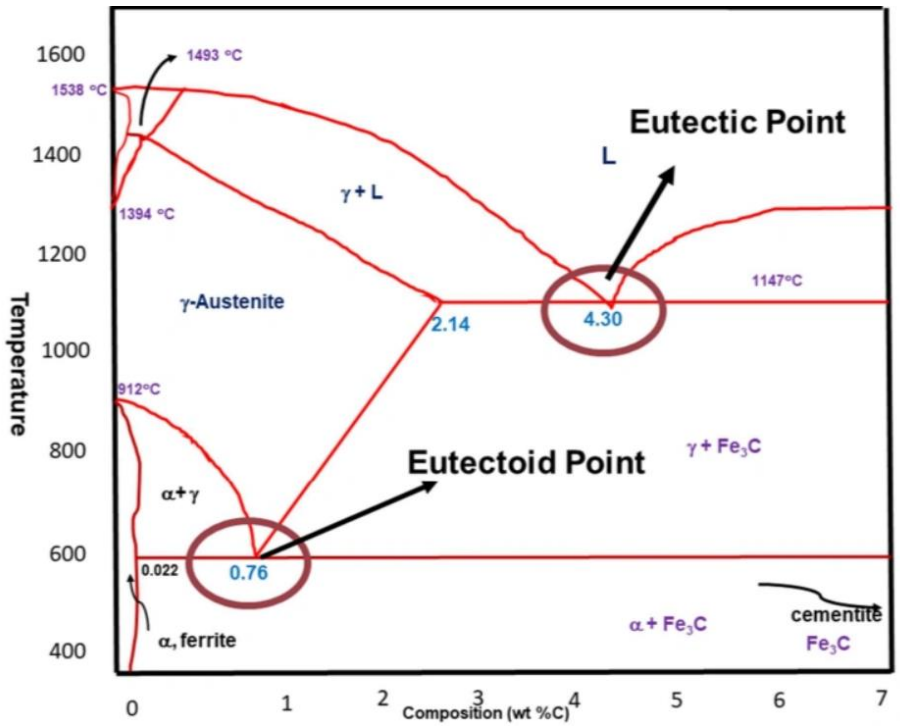
- When steel cools:
  - Austenite transforms into other phases.
- This is how factories control:
  - Hardness
  - Strength
  - Toughness
- Example:
  - Same steel composition can become soft or hard depending on heat treatment.

https://www.youtube.com/watch?v=SBoVR3jnucQ

Most Important Special Points (Only Mention)

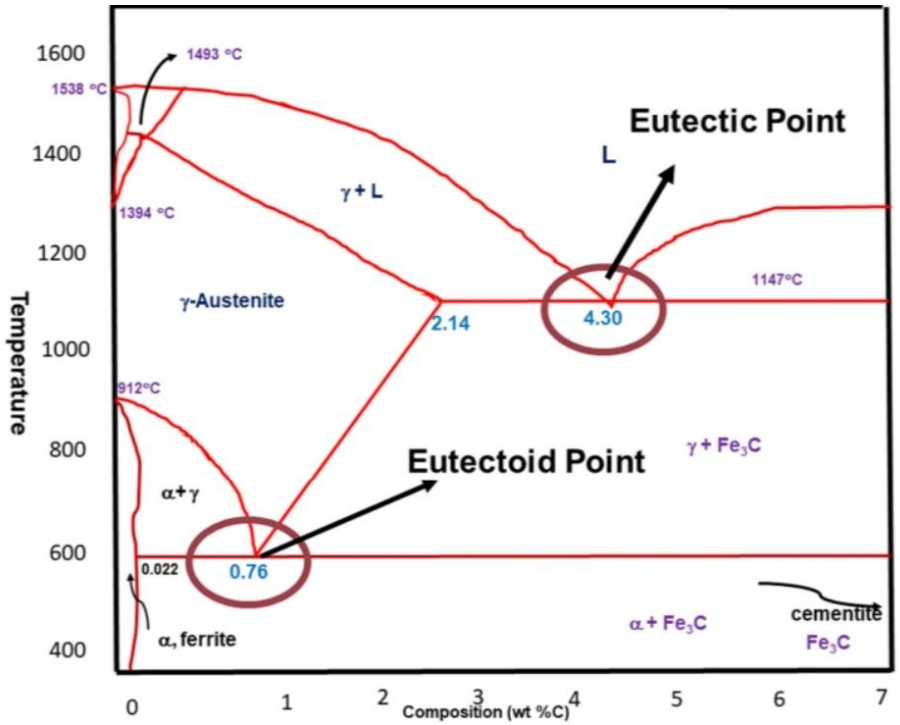
- Eutectoid Point (~0.8% C + 723°C)
  - Austenite → Pearlite
  - Very important in steel science.

- Eutectic Point (4.3% C + 1147°C)
  - Liquid → two solids
  - More related to cast iron.



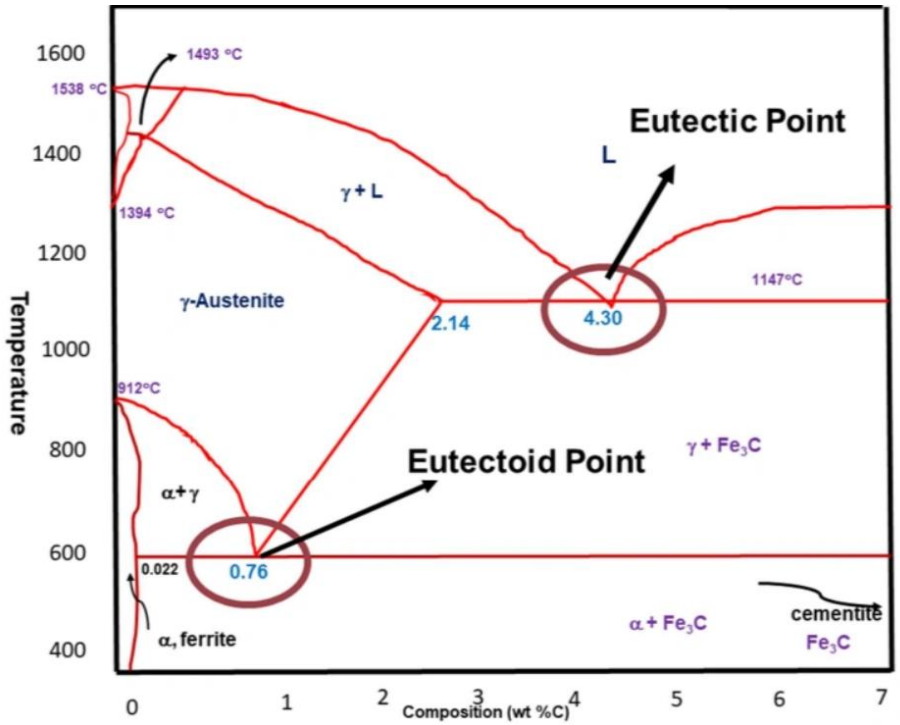
Eutectoid Point (Steel)

- Inputs:
  - 0.8% Carbon
  - 723°C
  - Material is **Austenite** (solid phase)
- Output after cooling:
  - **Pearlite** = Ferrite + Cementite
- Simple meaning:
  - One solid structure changes into **two new solid structures**.
- Why important:
  - Used to control steel strength and hardness.



Eutectic Point (Cast Iron)

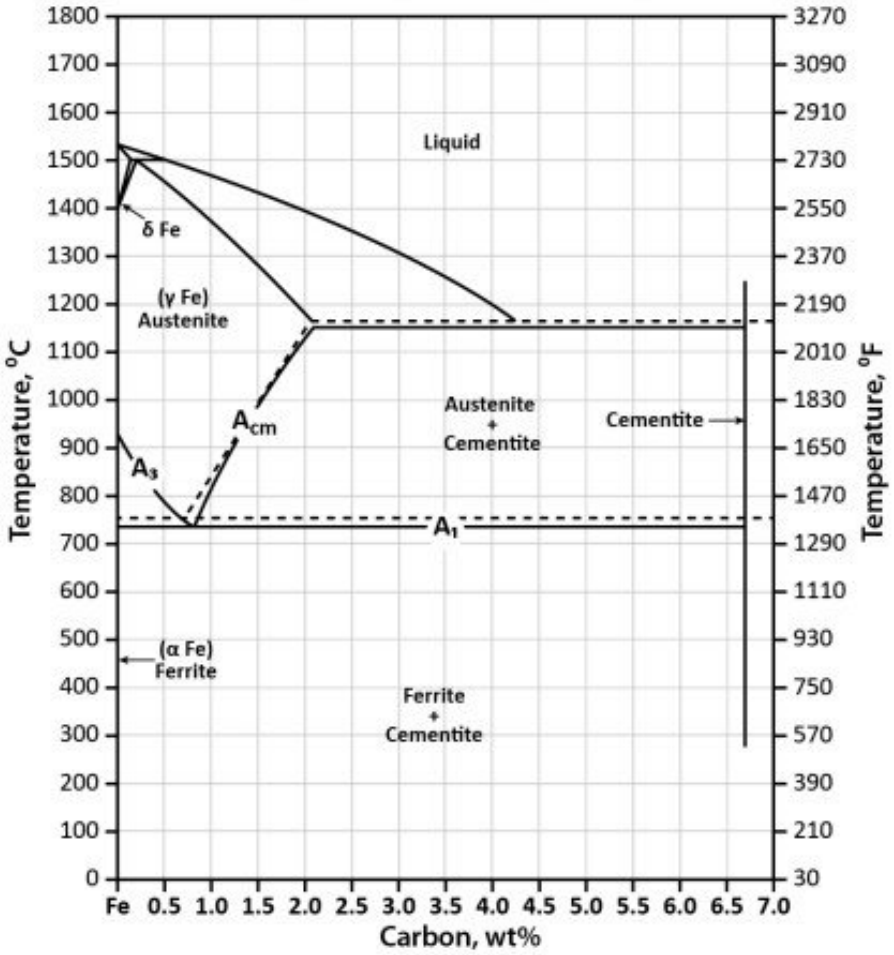
- Inputs:
  - 4.3% Carbon
  - 1147°C
  - Material is Liquid molten alloy
- Output after cooling:
  - Two solid phases form together
  - (usually Austenite + Cementite)
- Simple meaning:
  - Liquid metal solidifies into two solids at the sam time.
- Why important:
  - Very important in cast iron manufacturing and casting behavior.

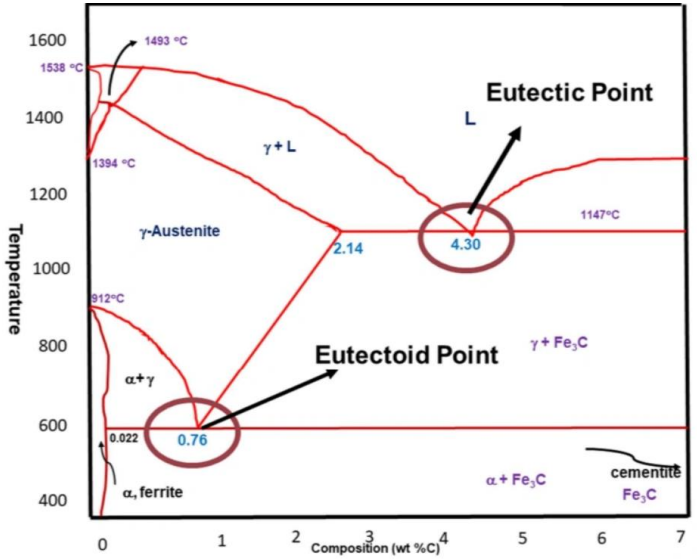


Eutectic -Eutectoid

- Eutectic
- Liquid → Two solids

- Eutectoid
- Solid → Two solids





Real Applications - Low Carbon Steel

- Carbon: ~0.05–0.25% C
  Typical processing temperatures: hot rolling ~900–1200°C
- **Applications:**
  - Construction beams
  - Car body panels
  - Pipes
- ***Why:*** ductile, weldable, cheap

Real Applications - Medium Carbon Steel

- Carbon: ~0.25–0.60% C
  Heat treatment: austenitize ~800–900°C then quench/temper
- Applications:
  - Shafts
  - Gears
  - Rails
  - Machine parts
- *Why:* stronger than low carbon steel

Real Applications- High Carbon Steel

- **Carbon:** ~0.60–1.0% (sometimes up to 1.4%)
  Heat treatment: ~780–900°C + hardening
- Applications:
  - Springs
  - Knives
  - Cutting tools
  - Wires
- **Why:** high hardness and wear resistance

Real Applications - Stainless Steel

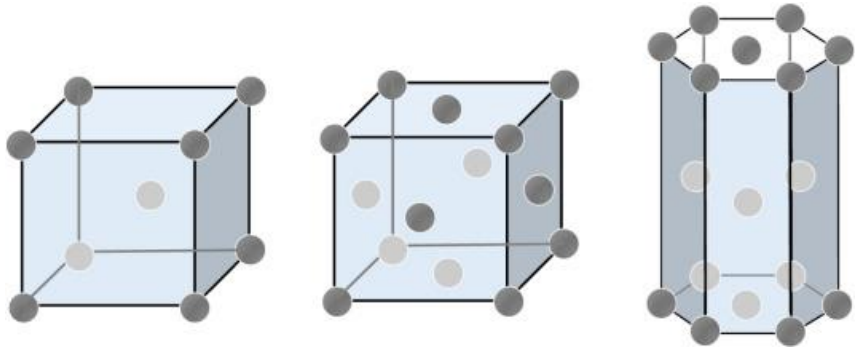
- **Carbon:** usually low (<0.08%)
  Chromium: >10.5% Cr (main key element)
- **Processing may use:**
  - Solution treatment ~1000–1100°C depending grade
- **Applications:**
  - Surgical instruments
  - Biomaterial implants (316L)
  - Food equipment
- *Why:* corrosion resistant + biocompatible

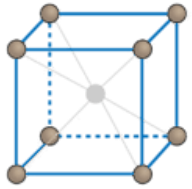
For Biomaterials →

- Stainless steel 316L used in implants
- Microstructure affects:
  - Strength
  - Corrosion resistance
  - Fatigue life
  - Biocompatibility

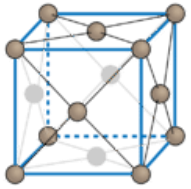


Crystal Structures in Biomaterials

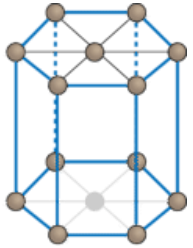




Cubic body centered (bcc)
Fe, V, Nb, Cr

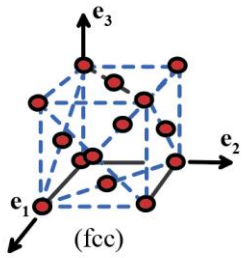


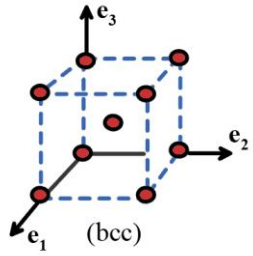
Cubic face centered (fcc)
Al, Ni, Ag, Cu, Au

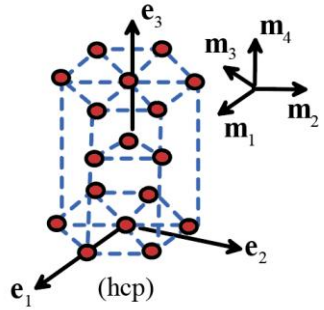


Hexagonal
Ti, Zn, Mg, Cd

Crystal Structures in Biomaterials







FCC — Face-Centered Cubic

BCC — Body-Centered Cubic

HCP — Hexagonal Close-Packed

Crystal Structures

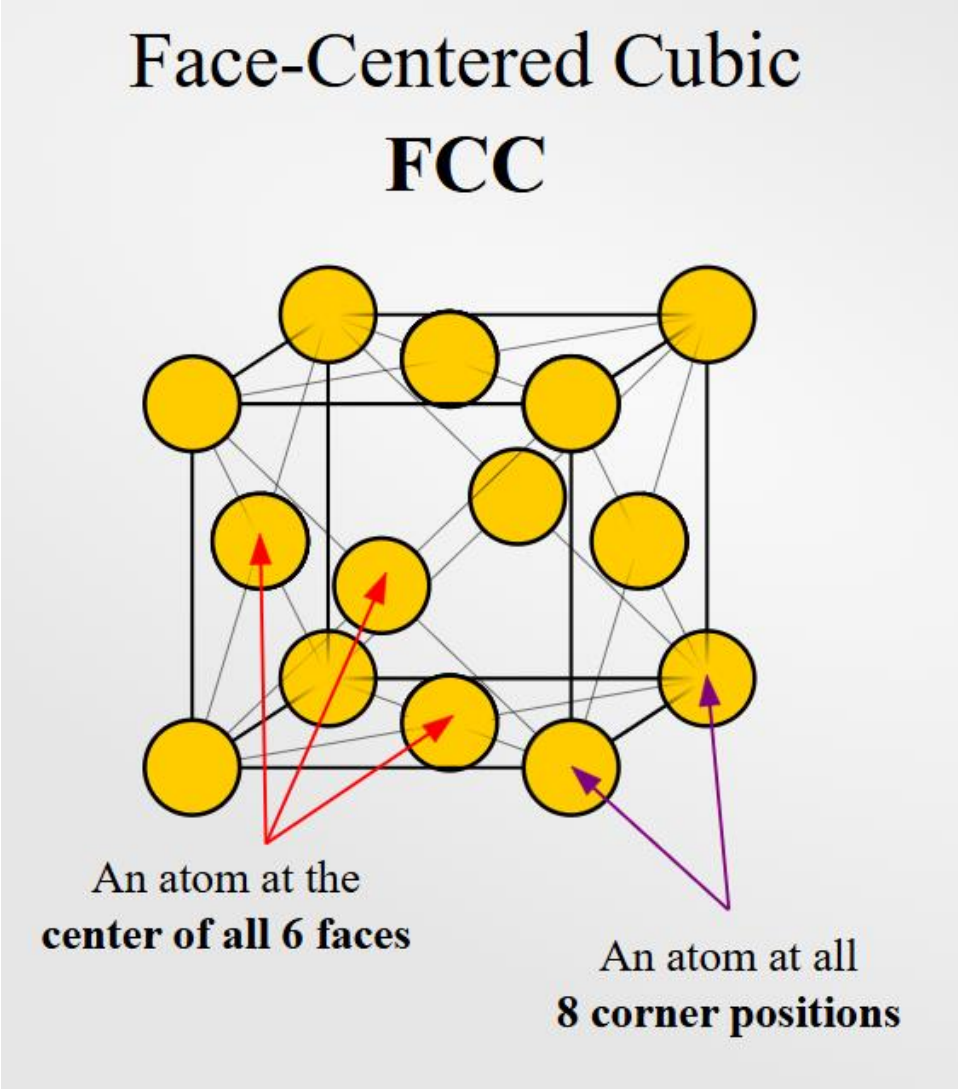
- Atoms inside metals are arranged in repeated patterns called crystal structures.
- Common structures:
  - FCC
  - BCC
  - HCP
- These structures affect:
  - Strength
  - Ductility
  - Brittleness
  - Implant performance

Why Crystal Structure Matters

- Crystal structure controls how easily atoms can move during deformation.
  - In biomaterials, this affects:
    - Mechanical strength
    - Fatigue resistance
    - Flexibility
    - Long-term implant stability

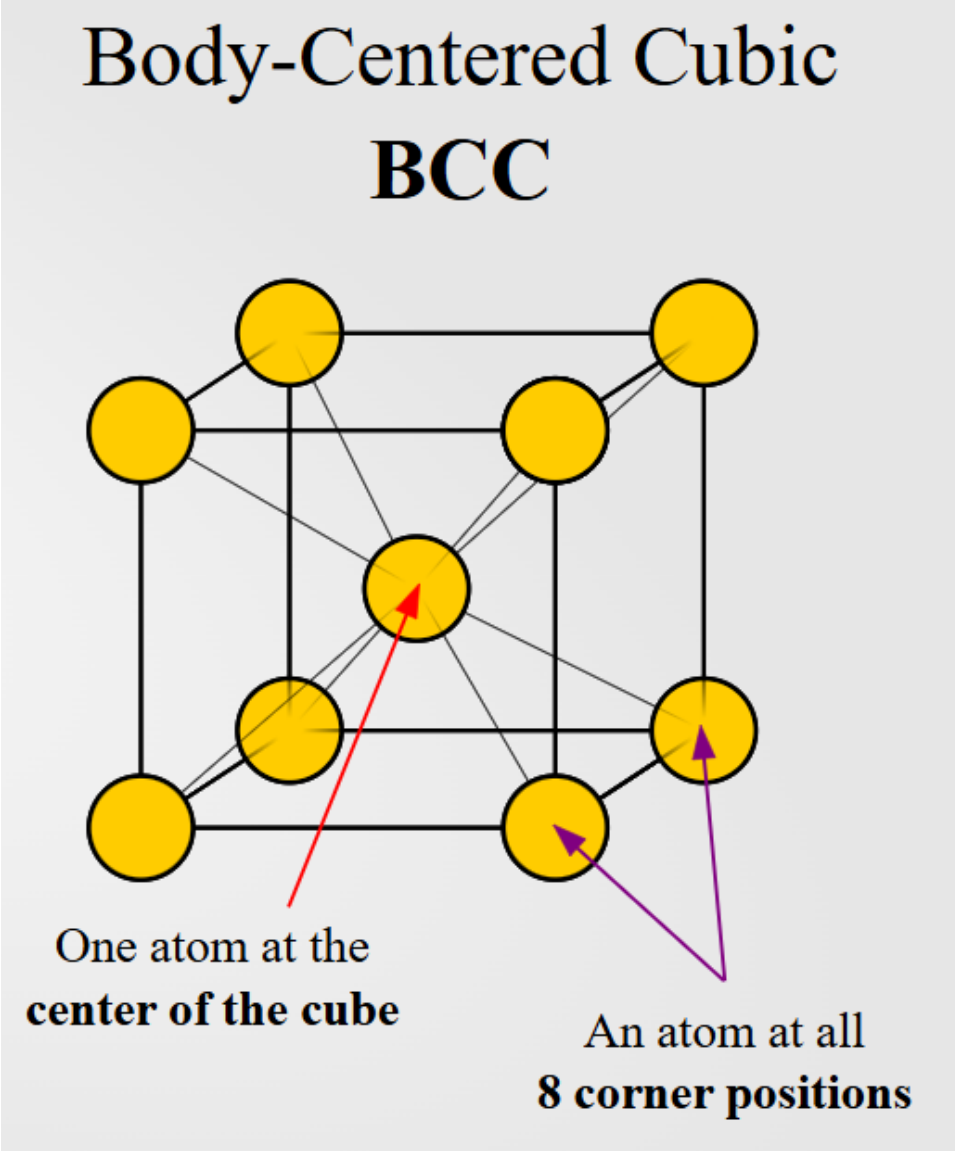
FCC Structure = Face-Centered Cubic

- Atoms are located:
  - At cube corners
  - At the center of each cube face
- Main properties:
  - High ductility
  - Good toughness
  - Easy to shape
- Examples:
  - Aluminum
  - Copper
  - Nickel
  - Gold
  - Silver
  - Austenitic stainless steel



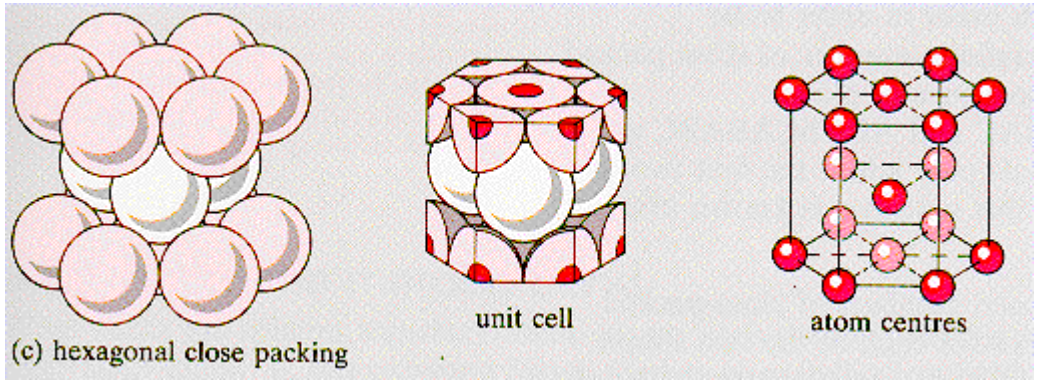
BCC Structure = Body-Centered Cubic

- Atoms are located:
  - At cube corners
  - One atom in the cube center
- Main properties:
  - Higher strength
  - Lower ductility than FCC
  - May become brittle at low temperature
- Examples:
  - Alpha iron
  - Chromium
  - Tungsten
  - Molybdenum
  - Vanadium



HCP Structure = Hexagonal Close-Packed

- Atoms are arranged in a hexagonal pattern.
- Main properties:
  - Strong
  - Less ductile
  - Limited deformation ability
- Examples:
  - Titanium
  - Magnesium
  - Zinc
  - Cobalt



Titanium (Struct.) Example in Biomaterials

- Titanium is important for:
  - Dental implants
  - Orthopedic implants
  - Joint replacements
- Titanium crystal structure changes with temperature:

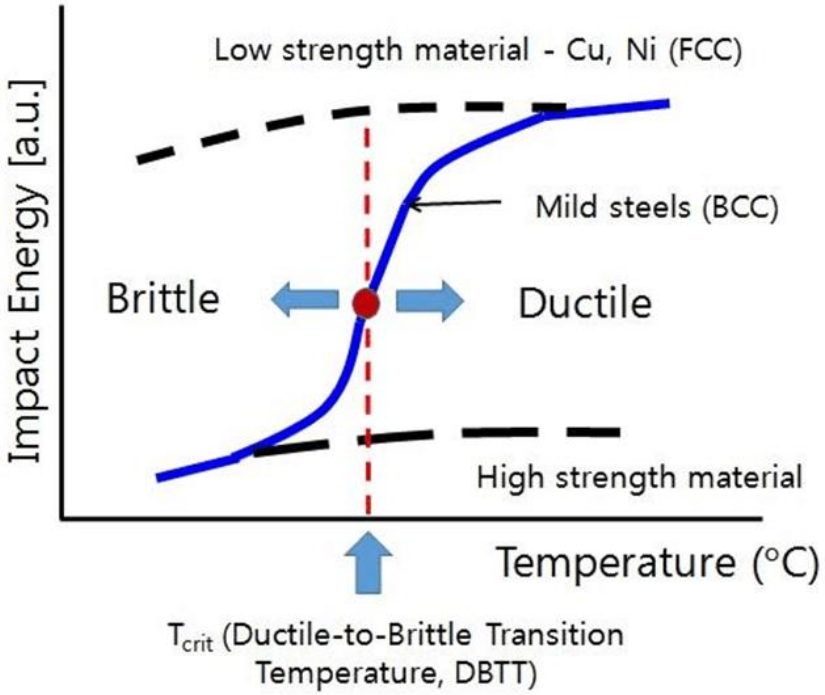
Temperature	Phase	Structure
Below 882.5°C	Alpha Ti	HCP
Above 882.5°C	Beta Ti	BCC

Simple Comparison

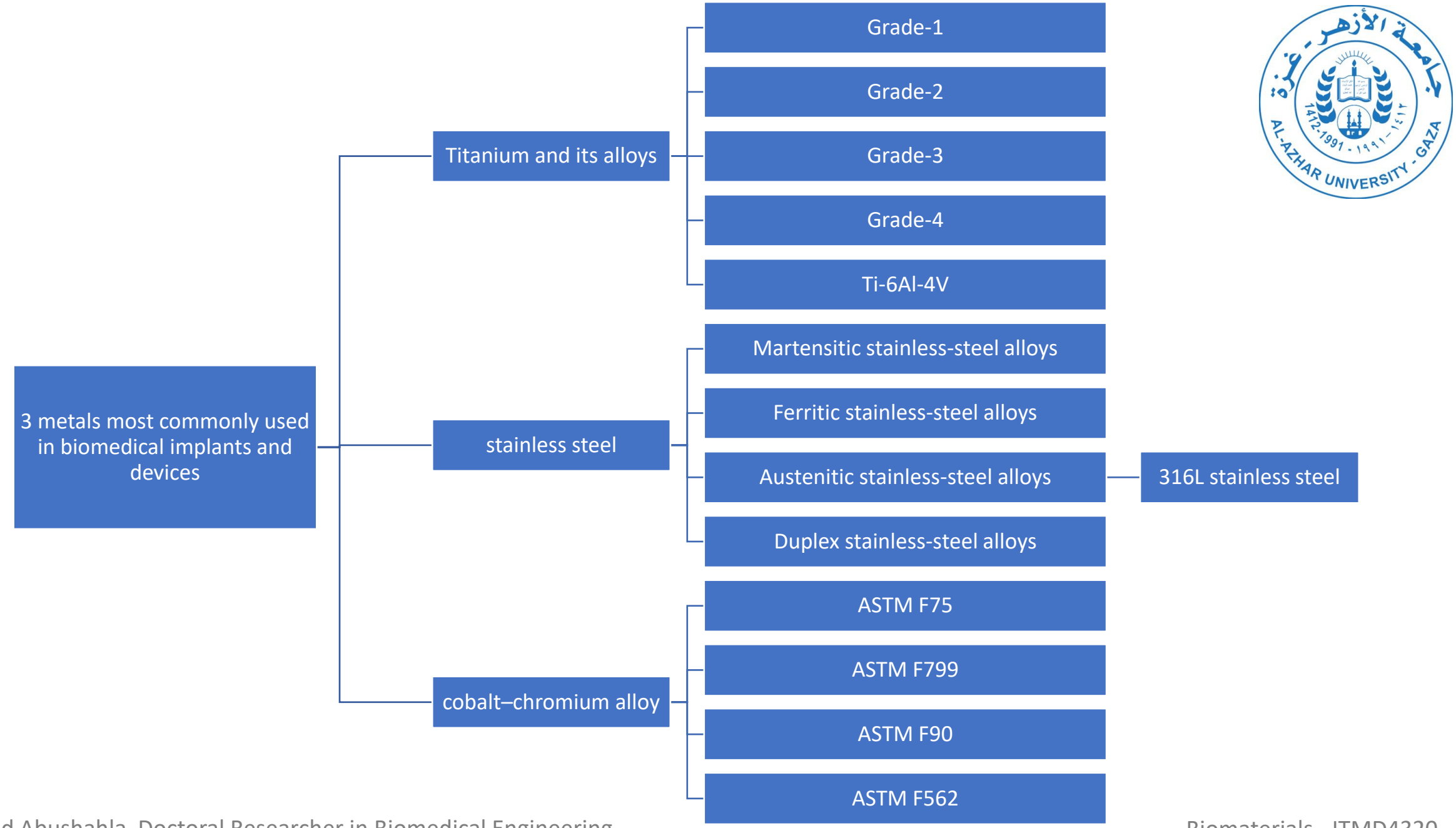
Structure	Main property	Biomedical relevance
FCC	Very ductile	Good formability
BCC	Stronger, less ductile	Structural metals
HCP	Strong but less ductile	Titanium implants

SO→

- Crystal structure is the atomic arrangement inside a material.
- **FCC:** ductile
  BCC: strong
  HCP: strong, less ductile
- For biomedical students, this is important because atomic structure affects how implant materials behave inside the body.



Metals: Structure & Properties


3 metals most commonly used in biomedical implants and devices
Titanium and its alloys
Grade-1
Grade-2
Grade-3
Grade-4
Ti-6Al-4V
stainless steel
Martensitic stainless-steel alloys
Ferritic stainless-steel alloys
Austenitic stainless-steel alloys
316L stainless steel
Duplex stainless-steel alloys
cobalt–chromium alloy
ASTM F75
ASTM F799
ASTM F90
ASTM F562

Metals as Biomedical Materials

- Metals are widely used in implants, medical devices, and surgical tools.
- Common applications:
  - ❑joint implants, bone plates/screws, dental implants, stents, heart valves, and pacemakers.
- Metallic biomaterials must have:
  - ❑corrosion resistance, biocompatibility, wear resistance, and good mechanical strength.
- A stable surface oxide layer helps protect metals from corrosion and supports biocompatibility.

Mechanical Requirements of Metallic Implants

- The metal's mechanical properties must match the body application.
- For bone implants, elastic modulus should be close to bone when possible.
- Very stiff metals can cause stress shielding, reducing load on bone.
- Stress shielding may lead to bone weakening or loss of bone quality.

Common Metals Used in Biomedical Implants

- The most common metallic biomaterials are **titanium alloys**, **stainless steel**, and **cobalt-chromium** alloys.
- Other examples include **nitinol**, **tantalum**, and **magnesium**.
- **Stainless steel** is **commonly** used in **coronary stents**.
- **Titanium** is **widely used** for **orthopedic** and **dental implants**.

1) Titanium

- Titanium is widely used because it is lightweight, corrosion resistant, and biocompatible.
- Its density is lower than stainless steel and cobalt–chromium alloys.
- Despite being lightweight, titanium has strong mechanical and chemical performance.
- Common uses: orthopedic implants, dental implants, and fixation devices.

Table 5.1a Chemical composition of four grades of cp-Ti and Ti-6Al-4V[a]

cp-Ti grades and Ti-6Al-4V	Elements					
	Carbon (% max)	Oxygen (% max)	Nitrogen (% max)	Hydrogen (% max)	Iron (% max)	Titanium (% max)
Grade-1	0.10	0.18	0.03	0.015	0.20	balance
Grade-2	0.10	0.25	0.03	0.015	0.30	balance
Grade-3	0.10	0.35	0.05	0.015	0.30	balance
Grade-4	0.10	0.40	0.05	0.015	0.50	balance
Ti-6Al-4V	0.08	0.13	0.05	0.0125	0.25	balance[b]

Titanium as a Biomaterial

- Lightweight, strong, corrosion-resistant, biocompatible
- Lower density than stainless steel and Co–Cr alloys
- Common uses: orthopedic implants, dental implants, fixation devices, pacemaker/VAD cases

Biomedical Titanium Types and Grades

- Main types: cp-Ti and Ti–6Al–4V
- cp-Ti grades 1–4 depend mainly on impurities: O, N, Fe
- More oxygen/impurities → higher strength, lower ductility
- Grade 1: more ductile; Grade 4: stronger
- Ti–6Al–4V: stronger and widely used

Crystal Structure and Alloy Classification

- Titanium is allotropic
- Alpha Ti: HCP, room temperature to 882.5°C
- Beta Ti: BCC, above 882.5°CAlloy classes: alpha, beta, alpha–beta
- Structure affects strength, formability, heat treatment, biomedical use

Titanium Alloy Families

Type	Stabilizers	Main properties	Biomedical note
Alpha	Al, Ga, Sn	Weldable, not heat-treatable	Limited use, lower strength
Beta	Mo, Ta, Nb, V	BCC, heat-treatable, formable	Important for new biomedical alloys
Alpha–Beta	Al + V	Strong, heat-treatable	Ti–6Al–4V, common in implants

Surface Properties, Modification, and Safety

- Titanium forms passive TiO_2 oxide layer: about 5–20 nm
- Gives corrosion resistance and supports biocompatibility
- Surface modification improves tissue response
- Methods: polishing, sandblasting, acid etching, anodization
- Ti–6Al–4V is common, but vanadium may cause concerns
- Alternatives: Ti–6Al–7Nb and newer beta titanium alloys

Biomedical Applications of Titanium Alloys

- Orthopedic implants: hip, knee, shoulder, and spinal implants
- Fracture fixation: bone plates, screws, rods, and nails
- Dental implants: dental roots, abutments, and orthodontic mini-screws
- Craniofacial implants: skull, jaw, and facial reconstruction plates
- Cardiovascular devices: pacemaker cases and ventricular assist device housings
- Main reason for use: high strength, low weight, corrosion resistance, and good biocompatibility

2) Stainless Steel

- Stainless steel is an iron-based alloy containing at least 10.5% chromium.
- Chromium forms a thin protective Cr_2O_3 oxide layer on the surface.
- This oxide layer gives stainless steel its corrosion resistance.
- Important alloying elements:
    - Cr: improves corrosion resistance
    - Mo: improves pitting corrosion resistance
    - Ni: improves formability and stabilizes austenite
    - N: increases strength and corrosion resistance
- Used in implants, medical devices, and surgical instruments.

Classification of Stainless Steel

- Stainless steels are classified mainly according to their microstructure.
- Microstructure affects strength, corrosion resistance, formability, and biomedical use.

Type	Main properties	Biomedical relevance
Martensitic	Strong, heat-treatable, moderate corrosion resistance	Surgical and dental instruments
Ferritic	Good corrosion resistance, limited mechanical strength	Limited biomedical use
Austenitic	Good corrosion resistance, good formability, good mechanical performance	Most important for implants
Duplex	Austenite + ferrite, high strength, good corrosion resistance	Future/advanced medical devices

316L Stainless Steel: Main Biomedical Grade

- 316L stainless steel is the most common stainless-steel grade for biomedical implants.
- It is mainly composed of Fe, Cr, Ni, and Mo.
- The letter “L” means low carbon content.
- Low carbon helps maintain corrosion resistance.316L has good:
  - Corrosion resistance
  - Formability
  - Mechanical strength
  - Clinical availability
  - Lower cost than titanium and Co–Cr alloys

Why 316L Works Well in Implants

- Chromium forms the protective oxide layer.
- Molybdenum improves resistance to pitting corrosion in body fluids.
- Low carbon reduces chromium carbide formation at grain boundaries.
- If chromium carbides form, less chromium remains for the protective oxide layer.
- Therefore, low carbon reduces corrosion risk.316L can also be strengthened by cold working.
- Cold working increases:
  - Yield strength
  - Tensile strength
  - Fatigue strength
- But cold working reduces ductility, so processing must be controlled carefully.

Biomedical Applications and New Developments

- Common biomedical uses of 316L stainless steel:
  - Bone plates
  - Screws
  - Fracture fixation devices
  - Coronary stents
  - Surgical instruments
  - Temporary orthopedic implants
- Duplex stainless steels contain both austenite + ferrite.
- They offer higher strength and good resistance to pitting and stress corrosion cracking.
- Their biomedical use is still more limited than 316L.

Biomedical Applications and New Developments

- Newer stainless-steel developments include:
  - Nitrogen-strengthened stainless steels for better strength and corrosion resistance
  - Nickel-free stainless steels to reduce allergy concerns in some patients
- So:316L is still the main biomedical stainless steel, but newer stainless steels aim to improve strength, corrosion resistance, and patient safety.

3) Cobalt–Chromium Alloys

- Co–Cr alloys are strong metallic biomaterials.
- They have excellent corrosion resistance and wear resistance.
- They are commonly used in load-bearing implants.
- Chromium forms a protective Cr_2O_3 oxide layer.
- Their final properties depend strongly on:
  - Chemical composition
  - Casting or wrought processing
  - Cold working / heat treatment

Common Biomedical Co–Cr Alloys

Alloy	Type	Main idea
ASTM F75	Cast Co–Cr–Mo	Good corrosion resistance, but casting defects may reduce strength
ASTM F799	Thermomechanically processed Co–Cr–Mo	Stronger than cast F75 because of forging/processing
ASTM F90	Wrought Co–Cr–W–Ni	Good fabrication and high strength after cold working
ASTM F562 / MP-35N	Wrought Co–Ni–Cr–Mo	Very high strength, ductility, and corrosion resistance

Processing Controls Co–Cr Performance

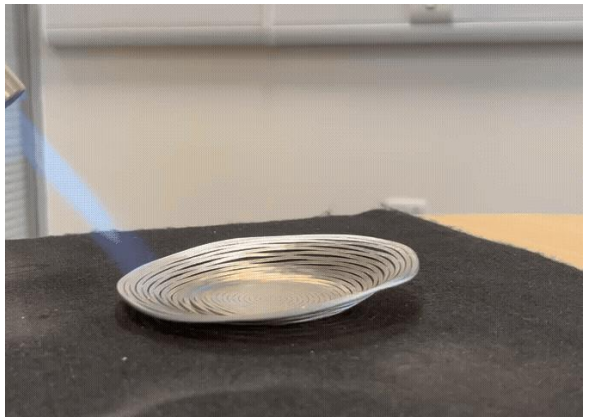
- Co–Cr alloy performance depends not only on composition, but also on processing.
- Casting is useful for complex shapes, but may produce large grains or defects.
- Forging / thermomechanical processing improves strength and reliability.
- Cold working can greatly increase strength.
- Heat treatment and precipitation hardening can further improve selected alloys such as MP-

Biomedical Importance

- Co–Cr alloys are selected when implants need:
  - High strength
  - High wear resistance
  - Good corrosion resistance
  - Long-term mechanical stability
- Common applications:
  - Hip and knee joint components
  - Dental implants/frameworks
  - Cardiovascular devices
  - High-strength fixation device

4) Nitinol

- Nitinol is a nickel–Titanium alloy.
- It has nearly equal amounts of nickel and titanium.
- It is a shape memory alloy.
- It can return to its original shape when heated.
- Important in smart biomedical devices.





In metallurgy, a shape-memory alloy (SMA) is an alloy that can be deformed when cold but returns to its pre-deformed ("remembered") shape when heated

Shape Memory Effect of Nitinol

- Nitinol can be deformed at low temperature.
- At higher temperature, it returns to its original shape.
- This behavior is caused by phase transformation.
- The two important phases are martensite and austenite.

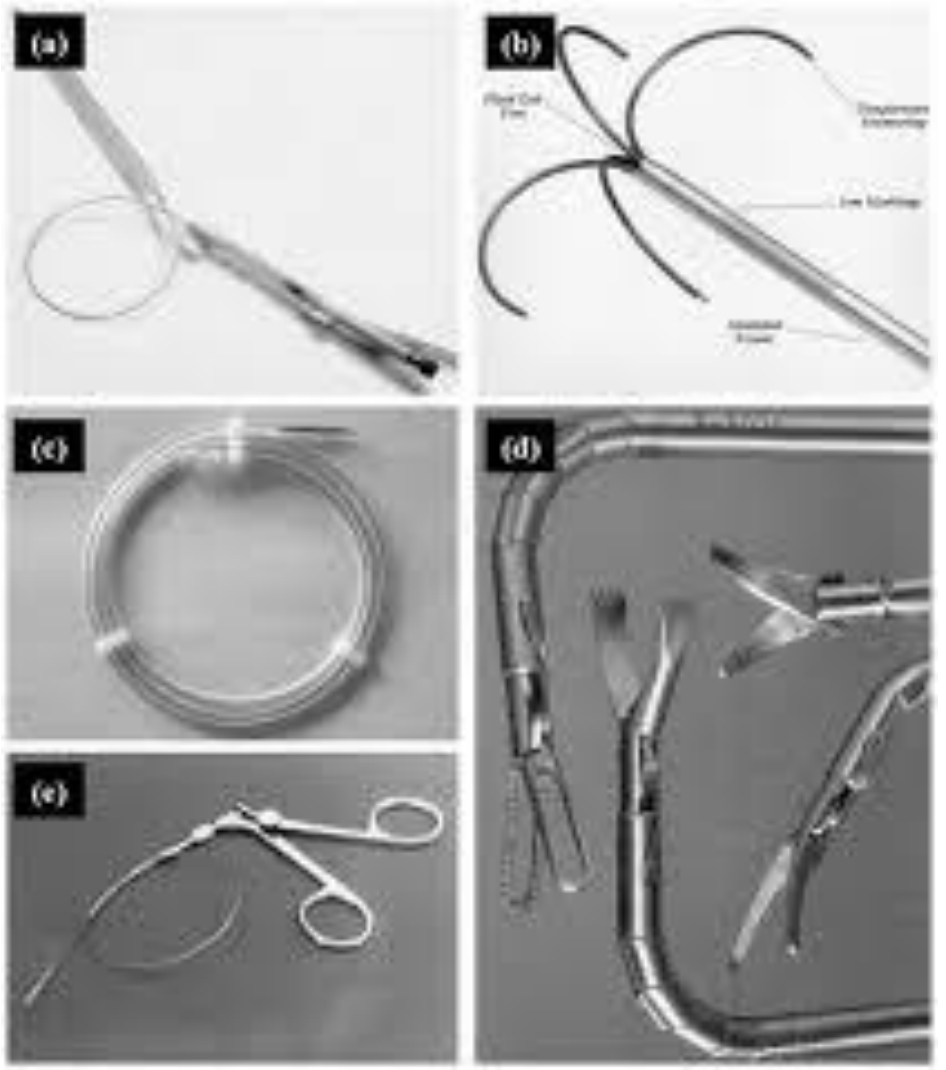
Biomedical Use of Shape Memory

- Nitinol is commonly used in self-expanding vascular stents.
- The stent is compressed before insertion.
- At body temperature, it expands to its memorized shape.
- This allows easier delivery through blood vessels.
- Useful for minimally invasive medical procedures.

Applications of Nitinol

- Self-expanding vascular stents.
- Orthodontic wires.
- Medical staples.
- Blood clot filters.
- Other flexible or shape-memory medical devices.





5) Tantalum

- Tantalum is used as a **pure metal**, an **alloying element**, or a **coating material**.
- It has excellent **biocompatibility** and **corrosion resistance**.
- Its good biological behavior comes from a stable **surface oxide layer**.
- Mechanically, tantalum has **lower strength** than titanium, stainless steel, and Co–Cr alloys.
- Therefore, it is not usually the first choice for major load-bearing implants.
- Biomedical applications include:
  - Radiographic markers
  - Coronary stents and vascular clips
  - Cranial defect repair
  - Fracture fixation
  - Dental implants
  - Biocompatible coatings

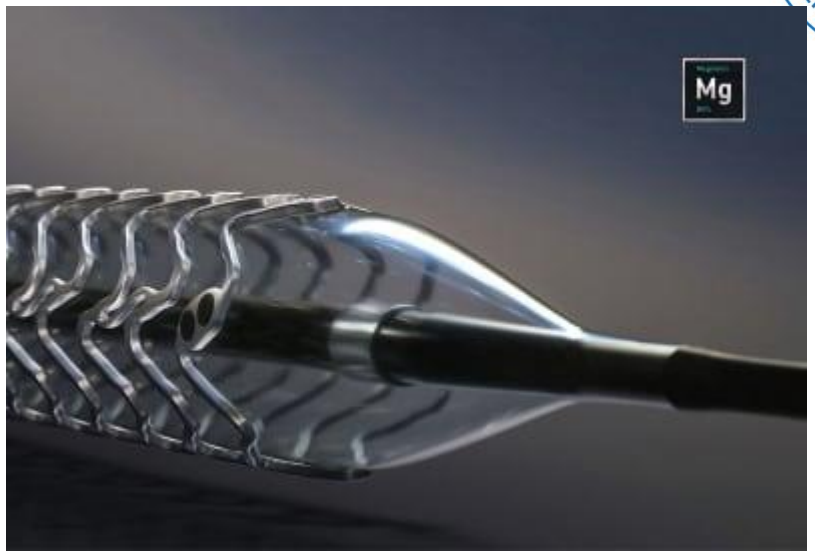


6) Magnesium

- Magnesium is lightweight and biodegradable.
- Its density and elastic modulus are closer to bone than many other metals.
- This may reduce stress shielding in orthopedic implants.
- Magnesium is naturally present in bone tissue.
- It can gradually degrade inside the body, which may reduce the need for a second removal surgery.

Applications of Magnesium

- Biomedical applications include:
  - Temporary orthopedic implants
  - Fracture fixation devices
  - Biodegradable coronary stents
  - Pediatric vascular stents



- → Magnesium is attractive because it is **bone-like and biodegradable**.

Summary of Metallic Biomaterials

- Titanium, stainless steel, and cobalt–chromium alloys are the most commonly used metallic biomaterials.
- Titanium alloys are lightweight, corrosion resistant, and biocompatible; common examples include cp-Ti and Ti–6Al–4V.
- Stainless steel, especially 316L, is widely used because of its good strength, corrosion resistance, and protective chromium oxide layer.
- Cobalt–chromium alloys provide excellent strength, wear resistance, and corrosion resistance, making them suitable for orthopedic and cardiovascular implants.

Summary of Metallic Biomaterials (Continued)

- **Nitinol** is a shape-memory Ni–Ti alloy widely used in self-expanding stents.
- **Tantalum** has excellent biocompatibility and corrosion resistance, but its mechanical strength is lower than many other implant metals.
- **Magnesium alloys** are promising biodegradable metallic biomaterials, especially for temporary implants, but corrosion control remains a major challenge.
- **Overall**, metals are important biomaterials because they provide strength, durability, and useful functional properties, while their performance depends strongly on corrosion behavior and biocompatibility.

Think with me 🤔🤔🤔

- Why are metals widely used in biomedical implants and medical devices?
- Why is corrosion resistance important for metallic biomaterials?
- Compare titanium, 316L stainless steel, and cobalt–chromium alloys: what is the main advantage of each one?
- How does a Nitinol self-expanding stent work, and why is it useful in medicine?



Thank you